Table 12.1
Quantities of Asphalt and Aggregate for Single-Surface Treatments

	Nominal Size of Aggregate	AASHTO Size No.	Quantity of [a] Aggregate lb/yd² (kg/m²)	Quantity of [b] Asphalt gal/yd² (l/m²)	Type and Grade of Asphalt
1	¾–½ in (19.0 mm–9.5 mm)	6	40–50 (22–27)	0.35–0.45 (1.6–2.0)	Asphalt cement [c]
				0.40–0.50 (1.8–2.3)	RS-2, CRS-2
2	½ in–No. 4 (12.5 mm–4.75 mm)	7	25–30 (14–16)	0.20–0.30 (0.9–1.4)	Asphalt cement [c]
				0.30–0.45 (1.4–2.0)	RS-1, RS-2 CRS-1, CRS-2
3	⅜ in–No. 8 (9.5 mm–2.36 mm)	8	20–25 (11–14)	0.15–0.25 (0.7–1.1)	Asphalt cement [c]
				0.20–0.35 (0.9–1.6)	RS-1, RS-2 CRS-1, CRS-2
4	No. 4–No. 16 (4.75 mm–1.18 mm)	9	15–20 (8–11)	0.15–0.20 (0.7–0.9)	RS-1, MS-1 CRS-1, HFMS-1
5	Sand	AASHTO M-6	10–15 (5–8)	0.10–0.15 (0.5–0.7)	RS-1, CRS-1 MS-1, HFMS-1

These quantities of asphalt cover the average range of conditions that include primed granular bases and old pavement surfaces. The quantities and types of materials may be varied according to local conditions and experience.

(a) The mass (weight of aggregate shown in the table is based on aggregate with a specific gravity of 2.65. In case the specific gravity of the aggregate used is lower than 2.55 or higher than 2.75, the amount shown in the table above should be multiplied by the ratio that the bulk specific gravity of the aggregate used bears to 2.65.

(b) The lower application rates of asphalt shown in the above table should be used for aggregate having gradations on the fine side of the specified limits. The higher application rates should be used for aggregate having gradations on the coarser side of the specified limits.

It is important to adjust the asphalt content for the condition of the road, increasing it if the road is absorbent and/or badly cracked, or decreasing it if the road is "fat" with flushed asphalt.

Corrections for Surface Conditions

Texture	gal/yd²	(l/m²)
Black, flushed asphalt	−0.01 to −0.6	−0.04 to −0.27
Smooth, nonporous	0.00	0.00
Absorbent		
– slightly porous, oxidized	+0.03	+0.14
– slightly pocked, porous, oxidized	+0.06	+0.27
– badly pocked, porous, oxidized	+0.09	+0.40

(c) AC-2.5, AC-5, AR-1000, AR-2000; 200/300 pen., 120/150 pen.
(*Note:* In some areas, persistent difficulty in retaining aggregate has been experienced with 200–300 penetration asphalt cements. Where this has occurred, the use of 200–300 penetration asphalt is not recommended.

In lieu of reiterating these design procedures, a range of asphalt and aggregate applications is provided in this section as a guide to estimating the proper quantity of materials for a given aggregate size. Adjustments should be made for actual field and material conditions. Tables 12.1, 12.2, and 12.3 provide some guidance on the proper asphalt and cover-aggregate application rates for single-, double-, and triple-surface treatments.

These quantities of asphalt cover the average range of conditions that include primed granular bases and old pavement surfaces. The quantities and types of materials may be varied according to local conditions and experience. The mass (weight) of aggregate shown in the tables is based on aggregate with a specific gravity of 2.65. In case the specific gravity of the aggregate used is lower than 2.55 or higher than 2.75, the amount shown in tables 12.1, 12.2, and 12.3 should be multiplied by the ratio that the bulk specific gravity of the aggregate used bears to 2.65. The lower application rates of asphalt shown in the tables should be used for aggregate having gradations on the fine side of the specified limits. The higher application rates should be used for aggregate having gradations on the coarser side of the specified limits.

It is important to adjust the asphalt content for the condition of the road, increasing it if the road is absorbent and/or badly cracked or decreasing it if the road is "fat" with flushed asphalt.

Table 12.2
Quantities of Asphalt and Aggregate per Square Yard (Square Meter) for Double-Surface Treatment

	Nominal Size of Aggregate	Quantity of Aggregate lb/yd² (kg/m²)	Quantity of Asphalt gal/yd² (l/m²)
½ in (12.5 mm) thick			
1st application *	⅜ in–No. 8 (9.5–2.36 mm)	25–35 (14–19)	0.20–0.30 (0.9–1.4)
2nd application	No. 4–No. 16 (4.75–1.18 mm)	10–15 (5–8)	0.30–0.40 (1.4–1.8)
⅝ in (15.9 mm) thick			
1st application *	⅜ in–No. 8 (12.5–4.75 mm)	30–40 (16–22)	0.30–0.4 (1.4–1.80)
2nd application	No. 4–No. 16 (4.75–1.18 mm)	15–20 (8–11)	0.40–0.50 (1.8–2.3)
¾ in (19.0 mm) thick			
1st application *	¾ in–⅜ in (19.0–9.5 mm)	40–50 (22–27)	0.35–0.50 (1.6–2.3)
2nd application	⅜ in–No. 8 (9.5–2.36 mm)	20–25 (11–14)	0.50–0.60 (2.3–2.7)

* If applied on untreated granular (aggregate) base, use cutback asphalt in lieu of emulsions.

Table 12.3
Quantities of Asphalt and Aggregate per Square Yard (Square Meter) for Triple-Surface Treatment

	Nominal Size of Aggregate	Quantity of Aggregate lb/yd² (kg/m²)	Quantity of Asphalt gal/yd2 (L/m²)
½ in (12.5 mm) thick			
1st application*	⅜ in–No. 8 (9.5–2.36 mm)	25–35 (14–19)	0.20–0.30 (0.9–1.4)
2nd application	No. 4–No. 16 (4.75–1.18 mm)	10–15 (5–8)	0.25–0.35 (1.1–1.6)
3rd application	No. 4–No. 100 (4.75 mm–150 μ m)	10–15 (5–8)	0.20–0.30 (0.9–1.4)
⅝ in (15.9 mm) thick			
1st application *	⅜ in–No. 8 (12.5–4.75 mm)	30–40 (16–22)	0.20–0.30 (0.9–1.4)
2nd application	⅜ in–No. 8 (9.5–2.36 mm)	15–20 (8–11)	0.30–0.40 (1.4–1.8)
3rd Application	No. 4–No. 16 (4.75–1.18 mm)	10–15 (5–8)	0.20–0.30 (0.9–1.4)
¾ in (19.0 mm) thick			
1st application*	¾ in–⅜ in (19.0–9.5 mm)	35–45 (19–25)	0.25–0.35 (1.1–1.6)
2nd application	⅜ in–No. 8 (9.5–2.36 mm)	20–30 (11–14)	0.30–0.40 (1.4–1.8)
3rd application	No. 4–No. 16 (4.75–1.18 mm)	10–15 (5–8)	0.25–0.35 (1.1–1.6)

* If applied on untreated granular (aggregate) base use cutback asphalt in lieu of emulsions.

Equipment

Distributor

The asphalt distributor is the most important piece of equipment on a surface treatment project. It is made specifically to apply the asphalt product uniformly and in specified quantities.

The asphalt distributor (see Figure 12.15) consists of a truck or trailer-mounted insulated tank ranging in capacity from 800 to 5500 gallons (3000–20,800 liters). Most distributors are equipped with a heating system that will maintain the asphalt at the proper spraying temperature.

The distributor has a power-driven pump capable of handling products ranging from light cold applications of emulsified asphalt to heavy asphalt cements heated to spraying viscosity temperatures. A system of spray bars with nozzles applies asphalt to the road's surface. These spray bars cover widths of 6–30 feet (1.8–9.1 meters) in one pass, depending on their width and the pump's capacity.

Although the methods of maintaining pressure may vary, all distributors use gear-type pumps to deliver asphalt to the spray bar. On some distributors, the pressure is governed by a variable-speed pump. On

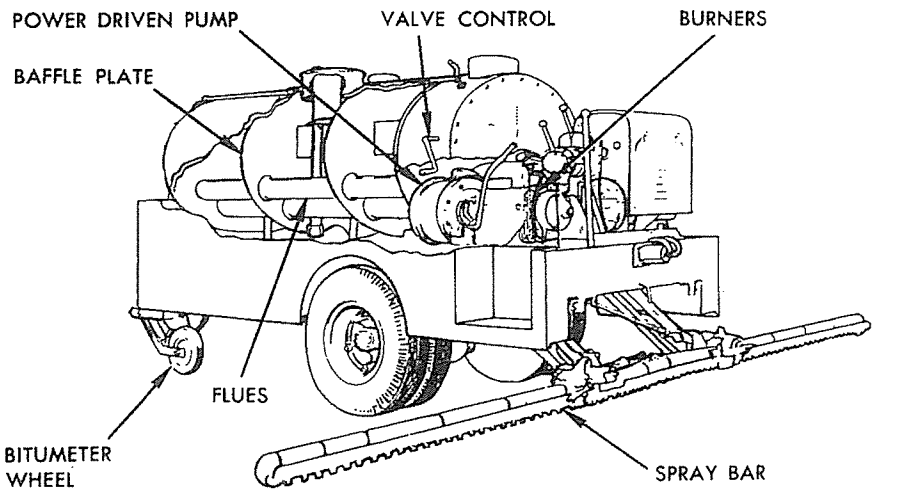


Figure 12.15 Asphalt Distributor

other distributors, it is controlled with a constant speed pump and a pressure-relief valve.

The correct pump speed or pressure is that which neither atomizes the asphalt nor distorts the spray fan. A low pressure may result in streaking from a nonuniform discharge of material from the individual nozzles. A high pressure, besides atomizing the asphalt, may distort the spray fan. Manufacturers supply charts and data for proper pump speed or pressure for determining the volume-per-minute discharge for each nozzle size.

A tachometer is used as an aid in maintaining uniform spreader speed. Newer distributors have interlocks between the asphalt pump and the forward speed of the distributor. As the distributor changes speed, the pump speed is adjusted to compensate for the change in speed of the distributor, thus maintaining a constant application rate.

Marking the length that each truckload of aggregate should cover can help monitor distribution rates.

The circulating system has an engine or hydraulic driven pump that

- fills the distributor tank;
- circulates material through the bar and tank;
- sprays material through the bar or hand spray;
- draws material back to the tank from the bar or hand spray;
- pumps material from the tank to outside storage; and
- transfers material from one storage tank to another.

One of the most important parts of the distributor is the spray bar. To achieve good results, the correct size nozzles for the type and grade of asphalt and the application rate must be selected. Before use, nozzles should be checked for damage and proper setting.

The angle of the long axis of the nozzle openings (see Figure 12.16) must be adjusted so that the spray fans do not interfere with each other. The nozzle angle varies according to the make of the distributor but is typically between

15 and 30 degrees (0.26 and 0.52 radians). It is important that all nozzles be set at the proper angle within close tolerances.

The height of the spray bar above the pavement surface must be adjusted to ensure uniformity of the asphalt spread. It is important, too, that the correct height be maintained throughout the application. A spray bar at the incorrect height causes streaking. Also, if the spray bar is set too high, wind distortion of the spray fans may occur.

For example, the best results with 4-inch (100-millimeter) nozzle spacing come from an exact triple coverage of the spray fans (see Figure 12.17). But in order to get triple coverage with 6-inch (150 millimeters) nozzle spacing, the height of the spray bar would be so great that it could be subject to wind distortion. In such cases, a double-coverage pattern is used.

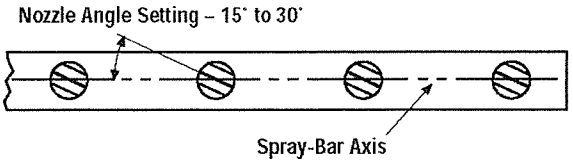


Figure 12.16 Proper Nozzle Angle Setting

Note. On occasion, some operators will set end nozzles at a different angle (60 to 90 degrees with respect to the spray-bar) in an attempt to obtain a good edge. This practice should NOT be permitted as it will produce a fat streak on the edge and rob the adjacent spray fan of the lap from this nozzle. A curtain on the end of the bar or a special end-nozzle with all nozzles set at the same angle will provide more uniform coverage and make a better edge.

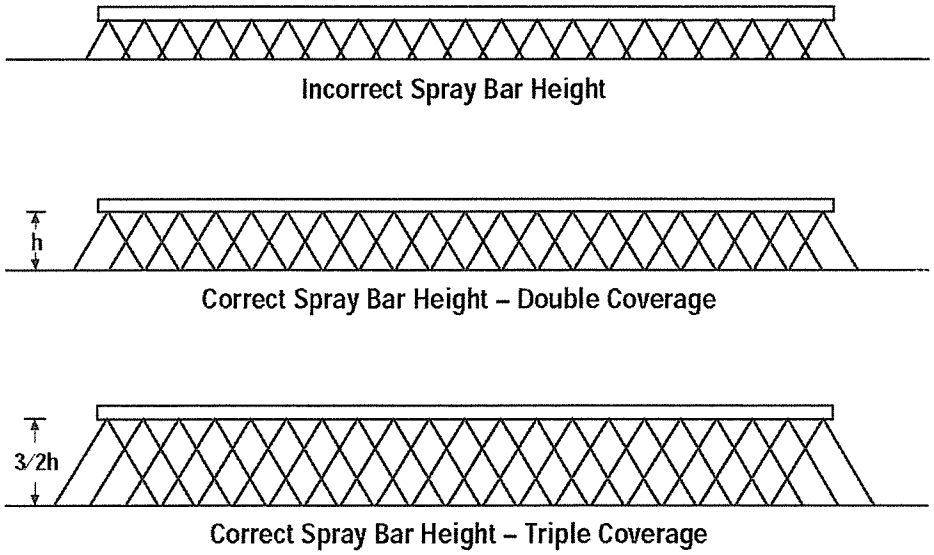


Figure 12.17 Overlapping Spray Fans

For best results, the height of the spray bar above the road surface should not vary more than ½ inch (13 millimeters) during application. Special steps must be taken to ensure this maximum permissible variance as the load lightens on the springs of the distributor truck. The frame of the distributor can be tied to the axle during spreading runs. Many distributors have mechanical controls to maintain the proper height.

Aggregate Spreader

The aggregate spreader is second in importance to the asphalt distributor. Its function is to apply a uniform aggregate cover at a specified rate over the freshly sprayed asphalt membrane.

Aggregate spreaders for use in surface-treatment construction consist of three basic types. One is the tailgate spreader, which is connected directly to the tailgate of the truck. Another is the mechanical spreader, which is also connected to the tailgate of the truck but has wheels to support the aggregate spreader. The third type is the mechanical, self-propelled spreader, which actually pulls the truck and supplies the power for the spreading operation rather than being dependent on the truck itself. There are several types of tailgate spreaders. The simplest is the vane spreader, as shown in Figure 12.18. It consists of a steel plate with a series of vanes, which spread the aggregate over the desired width on the road surface. There are tailgate spreaders that consist of a hopper with a feed roller, which is activated by small wheels that contact truck wheels (see Figure 12.19).

Truck-mounted mechanical spreaders are hoppers on wheels, which are hooked onto and are propelled by backing aggregate trucks. Hoppers usually contain augers to distribute the aggregate the full width of the box. The wheels support the spreader and operate the auger and also turn a scored or roughened feed roll in the bottom of the hopper. The feed roll ensures a uniform spread of material onto the road surface. There is an adjustable hitch on the front of the spreader by which it is attached to the aggregate truck. Mechanical spreaders also have controls to regulate the feed gates, the feed roll, the auger, and the truck hitch height.

A self-propelled spreader is shown in Figure 12.20. This machine provides a very uniform and continuous distribution of the aggregate, and

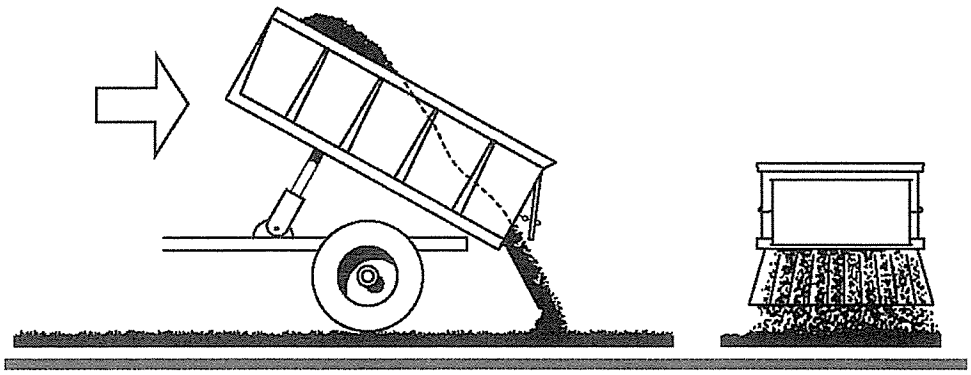


Figure 12.18 Tailgate Vane Spreader

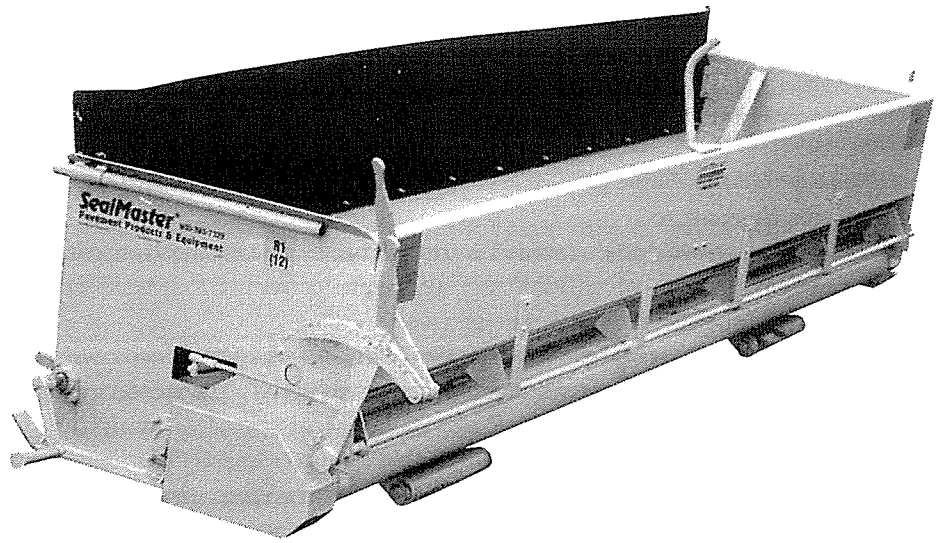


Figure 12.19 Truck-Attached Mechanical Spreader

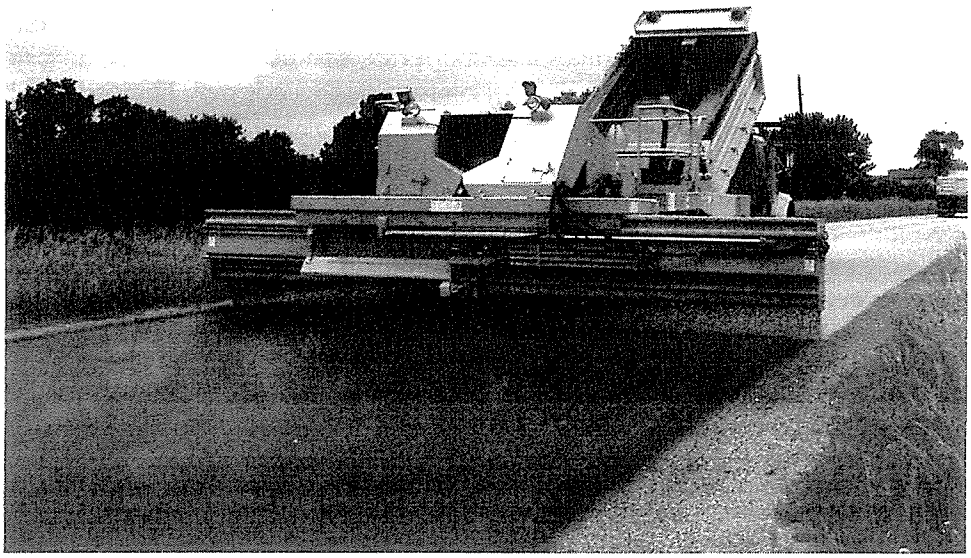


Figure 12.20 Self-Propelled Mechanical Spreader

makes very high production rates possible. The self-propelled spreader is connected to the aggregate truck by means of a self-locking hook, which locks on to the truck as the truck backs into the rear of the machine. The spreader then pulls the truck along the spread area. A receiving hopper is located at the rear of the machine. The material is moved from the receiving hopper to the spreading hopper by means of two individual—usually automatically controlled—conveyor belts.

By adjusting the speed of each conveyor belt, the flow of aggregate into the hopper can be set to match the flow from the hopper. Material-level sensors in the spread hopper, combined with belt speed control, assure the operator of a consistently full-spread hopper without a second person on the machine.

With individually controlled gates, the operator can utilize as little as 6 inches (150 millimeters) of the width of the spreader to as much as the complete width of the spreader without leaving the operator's seat or having a second person on the machine. This facilitates constructing radii and intersections and variable roadbed widths. The material, when dropping into the spreading hopper, usually passes through a scalping screen at the top of the hopper. This allows the scalping of any oversize material that may get into the receiving hopper from the truck. Material drops down into the spreading hopper and flows over a spread roll. The aggregate particles are then deposited onto the asphalt-coated surface.

The calibration and adjustment of all types of aggregate spreaders should be made according to manufacturers' instructions and operation manuals. Some additional suggestions to ensure good results:

- A tachometer used as an aid in maintaining uniform spreader-speed has proven to be highly successful.
- Distribution rates can be closely monitored by marking the length that each truckload of aggregate should cover.
- A quick check on the amount of aggregate being applied (rate of application) can be made by laying a section of cloth or building paper that measures 1 square yard (1 square meter) on the pavement—or by supporting a shallow box measuring 1 square yard (1 square meter) above the asphalt with nails or screws—and by passing over it with the spreader. The cloth, paper, or box is then carefully lifted, and the aggregate is weighed. This gives the application rate of the aggregate being spread.

Rollers

The seating of aggregate particles is an important part of the surface treatment operation. There are several types of compactors, but pneumatic-tire rollers are normally used.

A self-propelled pneumatic-tire roller (Figure 12.21) with tire pressures in the range of 60–90 pounds per square inch (415–620 kiloPascals) is recommended for use on asphalt-aggregate surface treatments. The tires force the aggregate firmly into the asphalt binder and to the flat side without crushing the particles. Steel-wheel rollers bridge the smaller-size particles and small depressions in the surface. Steel-wheel rollers also might crush the softer particles so that degradation takes place even before traffic uses the new surface.



Figure 12.21 Pneumatic-Tire Roller

Cleaning Equipment

Unless the surface to be covered is completely clean, the asphalt may not adhere to the pavement. It is therefore necessary to clean the surface before spraying the asphalt emulsion. A power street sweeper is recommended for picking up both dust and loose particles; if one is not available, a rotary power broom can be used. When brooms are used, flushing with water may be necessary to meet clean air standards or to remove caked-on materials.

Power sweepers or brooms are also used to remove loose particles after the treatment is completed and the asphalt has properly cured. Only light broom pressure should be used so that aggregate particles embedded in the asphalt membrane are not dislodged.

Construction Sequence

The following steps illustrate the sequence of operations for a chip seal:

1. Prepare the surface to be treated. Refer to section 12.4.1 in this chapter.
2. Apply the asphalt binder.

   Because the asphalt is sprayed on the roadway during construction of surface treatments, the temperature giving the most effective viscosity should be selected. The temperatures shown in Table 12.4 should be used as a guide. Each distributor has one or more thermometers for checking the asphalt temperature. The temperature used for spraying at a given spray-bar pressure should be that at which no fogging occurs when the asphalt material leaves the spray nozzle.

Table 12.4
Suggested Temperatures for Spraying Asphalt

Type and Grade of Asphalt	Spraying Temperature of Asphalt	
	Degrees Fahrenheit	Degrees Celsius[1]
Emulsified asphalts		
CRS-1	125–185	50–85
CRS-2	125–185	50–85
RS-1	70–140	20–60
RS-2	125–185	50–85
MS-1	70–160	20–70
HFMS-1	70–160	20–70
Asphalt cements	265+[2]	130+[2]

Notes:

[1] Exact conversions for °F are rounded to the nearest 5°C.

[2] The maximum temperature shall be below that at which fogging occurs.

The speed of the distributor and the length of the asphalt shot should be determined before the application starts; distributor speed is determined by

Customary **Metric**

$$S=\frac{9Q}{WR} \qquad S=\frac{Q}{WR}$$

Where

S = distributor speed, feet/minute (meters/minute)
Q = spray bar output, gallons/minute (liters/minute)
W = spray width, feet (meters)
R = rate of application, gallons/yard^2 (liters/meter^2)

Asphalt should be at the proper temperature for spraying. If it is, application at the correct rate presents no problem. The driver merely maintains the predetermined speed as indicated on the dial of the bitumeter.

The length of asphalt shot is determined before beginning each asphalt application. It is based on the number of loaded aggregate trucks on hand when operations begin. No more asphalt is applied than can be covered with aggregate within one minute. The length of spread may be determined by

Customary **Metric**

$$L=\frac{9T}{WR} \qquad L=\frac{T}{WR}$$

Where

L = length of spread, feet (meters)
T = total gallons (liters) to be applied from the distributor to the surface

W = spray width, feet (meters)
R = rate of application, gallons/yard^2 (liters/meter^2)

Once the length of the shot has been determined, the distributor driver merely has to maintain a preset speed. Although the distributor contains several metering devices to indicate asphalt application rate, it is necessary to make specific calculations after each shot. This can be done quite simply using the formula that follows. To check the rate of application, it is necessary to know the number of gallons (liters) of asphalt sprayed as well as the length and width of the spread.

Customary	**Metric**
$R=\frac{9TM}{WL}$	$R=\frac{TM}{WL}$

Where
R = rate of application, gallons/yard^2 (liters/meter^2)
T = total liters (gallons) spread from the distributor at spraying temperature (gauge stick reading before spread minus gauge stick reading after spread)
M = multiplier for correcting asphalt volume to basis 60°F (15.6°C), from temperature-volume tables
W = width of spread, feet (meters)
L = length of spread, feet (meters)

3. Apply the aggregate.

All aggregate needed for the planned spread should be on hand before starting. When the distributor moves forward to spray asphalt, the aggregate spreader should follow immediately behind it. The asphalt must be covered within one minute. Otherwise, the increase in viscosity that takes place within that time may prevent good wetting and binding of the aggregate. It is also important that aggregate be spread uniformly at the specified rate. In a single application, aggregate normally does not stick to the asphalt if applied more than one particle thick. It is useless and wasteful to apply the aggregate at a rate greater than for a single layer.

Excellent control is possible with mechanical spreaders. A uniform application rate can be ensured with a properly adjusted spreader if a tachometer is used to maintain uniform speed. Another aid in controlling distribution rates is to mark the length that each truckload of aggregate should cover.

4. Roll aggregate into the asphalt binder.

Rolling seats the aggregate into the asphalt membrane and thus promotes the bond necessary to resist traffic stresses. Pneumatic-tire rollers should be used on all surface treatment jobs. Pneumatic-tire rollers give uniform pressure over the entire area. Better results will be obtained if two self-propelled rollers are used with each aggregate spreader.

Pneumatic-tire rolling begins immediately after the cover material has been distributed and continues until the aggregate is properly

seated in the asphalt membrane. As soon as the asphalt has a definite set or hardening, rolling is discontinued to prevent the bond between the surface and the aggregate from being broken by the roller. Rolling begins at the outer edge of the treatment and proceeds in a longitudinal direction, working toward the center of the road. Each trip should overlap the previous trip by about one-half the width of the front wheels.

5. Remove excess aggregate.

   Excess aggregate, if placed, should be removed immediately with square-end shovels. In areas where application is insufficient, additional aggregate should be added immediately. However, with properly adjusted and operated aggregate spreaders, handwork should not be necessary. Also, excess aggregate under the action of traffic can loosen embedded aggregate.

   Aggregate that does not stick is often picked up by the tires of fast-moving vehicles and then thrown against following vehicles, potentially damaging headlights, windshields, and vehicle paint. Loose aggregate may be removed by light brooming with a rotary power broom. This brooming should be delayed until the asphalt has set and a good bond has developed. The cool of early morning is a good time for brooming.

   Despite precautions, some loose aggregate is usually left on the road surface after rolling is completed. Before the adjacent lane is covered with asphalt, loose aggregate is swept from along the joint and, if necessary, from the rest of the uncovered lane.

6. Control traffic.

   Traffic control is important to high-quality work, and it must be maintained throughout the job. High-speed traffic over a fresh surface treatment displaces aggregate and produces a slick, black surface. Traffic should be detoured or allowed only in the lane not under construction. When work is completed and initial asphalt set has occurred, traffic is still maintained at less than 20 miles per hour (32 kilometers/hour) until the asphalt finally sets. The time until final setting varies with the weather. Traffic must be controlled in a manner that provides maximum safety for the workers and the least possible interruption of the work. Warning signs, flaggers, and a pilot truck leading vehicles past the work area are safe and efficient means of controlling traffic.

   Traffic control also extends to the hauling equipment. Aggregate hauling trucks are routed to the aggregate spreader in a direction opposite to the progress of the surface treatment operation. This prevents trucks from being turned on the freshly placed treatment. All trucks are required to turn around at a designated spot away from the new work.

   If a double- or triple-surface treatment is required, steps 2 through 6 should be repeated.

   There should be a curing period between each treatment. The preceding treatment should be allowed to cure, under favorable conditions, for a period of one to seven days before placing the subsequent application of asphalt.

Seasonal Limitations

Weather plays an important role in the success of surface treatments. The best time for a surface treatment program is when weather conditions are most likely to be hot and dry during—and for some weeks after—treatment.

Many specifications require the air temperature to be at least 50°F (10°C) before surface treatment operations begin. Some require the road surface temperature to be above 70°F (20°C) before work starts. No matter how hot the asphalt is when sprayed, it will cool to the temperature of the pavement surface in one minute or less.

Surface treatment should never be started when the surface is wet or when rain threatens. The combination of water, fresh surface treatment, and traffic will result in loss of cover aggregate.

Transverse Joints

Rough and unsightly transverse joints can be avoided by starting and stopping the asphalt and aggregate spread on building paper. The paper is placed across the lane to be treated so that the forward edge is at the desired joint location. The distributor—traveling at the correct speed for the desired application rate—starts spraying on the paper so that the spray bar makes a full, uniform application when reaching the exposed surface. A second length of building paper is placed across the lane at the predetermined cutoff point for the distributor. This procedure gives a straight, sharp transverse joint. After the aggregate spreader has passed over it, the paper should be removed and disposed of properly.

For the next application, the leading edge of the paper is placed within a ½ inch (13 millimeters) of the cutoff line of the previously laid treatment. This prevents a gap between the two spreads.

Longitudinal Joints

Full-width applications of asphalt and aggregate eliminate longitudinal joints. However, in most surface treatment work, the joint is unavoidable because traffic lanes must be maintained.

To prevent aggregate from building up on the longitudinal joint, the edge of the aggregate spread should coincide with the edge of the full thickness of applied asphalt. The asphalt applied at full thickness is narrower than the total width of applied asphalt by one-half the width of coverage of the nozzles on the end of the spray bar. This leaves a width in which asphalt is present in partial thickness—and it will be overlapped when asphalt is applied in the adjacent lane. Then, when the aggregate is spread for the full width in the next lane, there is no buildup at the joint. The width of the asphalt strip left exposed varies—typically 6–8 inches (150–200 millimeters)—depending on the nozzle spacing and whether it is a triple- or double-lap spray pattern.

If possible, the longitudinal joint should be along the centerline of the pavement being treated. An established line helps to ensure a straight longitudinal joint.

12.4.4 Slurry Seals

A slurry seal is a mixture of well-graded fine aggregate, mineral filler (if needed), emulsified asphalt, and water applied to a pavement as a surface treatment. It is used in both the preventive and corrective maintenance of asphalt pavement surfaces. It does not increase the structural strength of a pavement section. Any pavement that is structurally weak in localized areas should be repaired before applying the slurry seal. All ruts, humps, low pavement edges, crown deficiencies, waves, or other surface irregularities that diminish the riding quality should be corrected before placing the slurry seal.

Slurry seal, when properly applied to the surface of a pavement, can be very effective. Its timely application can help reduce surface distress caused by oxidation of the asphalt and the embrittling of the paving mixture. It will seal the surface cracks, stop raveling and loss of matrix, make open surfaces impermeable to air and water, and improve skid resistance and pavement appearance.

Slurry seal has a number of advantages, including

- rapid application;
- no loose cover aggregate;
- excellent surface texture for paint striping;
- the ability to correct minor surface irregularities;
- minimum loss of curb height;
- no need for manhole or other structure adjustments; and
- improving the aesthetics of the existing pavement.

In many cases, the relatively low cost of the treatment makes it practical to import aggregates for special effects, such as high skid resistance, color contrast, and noise reduction.

Materials

Aggregates

The aggregate used in slurry seal must be clean, angular, durable, well-graded, and uniform. An individual aggregate or a blend of aggregates to be used in a slurry mix should meet these limits:

- Sand equivalent value, ASTM D 2419 (AASHTO T 176) = 45 minimum
- Los Angeles abrasion loss, ASTM C 131 (AASHTO 6) Grading C or D = 35 maximum
- The amount of smooth-textured sand of less than 1.25 percent water absorption is limited to not more than 50 percent of the total combined aggregate.

The three generally accepted gradings used for slurry mixtures are shown in Table 12.5.

Type I is used for maximum crack penetration. Also, it makes an excellent pretreatment for an HMA overlay or chip seal. It is usually used in low-density traffic areas such as light-aircraft airfields, parking areas, or shoulders where the primary objective is sealing.

Type II is the most widely used gradation. It is used to seal; to correct severe raveling, oxidation, and loss of matrix; and to improve skid resistance.

Table 12.5
Slurry Mixture Gradings

Type of Slurry	I	II	III
General usage	Crack filling and fine seal	General seal, medium-textured surfaces	1st and/or 2nd application, two-course slurry, highly textured surfaces
Sieve Size	Percent Passing		
⅜ in (9.5 mm)	100	100	100
No. 4 (4.75 mm)	100	90–100	70–90
No. 8 (2.36 mm)	90–100	65–90	45–70
No. 16 (1.18 mm)	65–90	45–70	28–50
No. 30 (600 μ m)	40–65	30–50	19–34
No. 50 (300 μ m)	25–42	18–30	12–25
No. 100 (150 μ m)	15–30	10–21	7–18
No. 200 (75 μ m)	10–20	5–15	5–15
Residual Asphalt Content, % Weight of dry aggregate	10–16	7.5–13.5	6.5–12
Application Rate, lb/yd^2 (kg/m^2), based on mass (weight) of dry aggregate	6–10 (3–5.5)	10–15 (5.5–8)	15 (8) or more

Recommended by International Slurry Surfacing Association

It is used for moderate traffic, depending on the quality of aggregates available and the design.

Type III is used to correct surface conditions, as the first course in multicourse applications for heavier traffic, and to provide skid resistance.

Asphalt

Emulsified asphalt used in the slurry mix may be SS-1, SS-1h, CS-1, or CSS-1h. Quick-setting (QS) asphalt emulsion is being used when an early opening to traffic is necessary. Sometimes, a small amount of liquid or powdered additive is added to the asphalt emulsion to control the setting time of the slurry mixture. This additive starts the set in anionic quickset emulsions. It retards the set in cationic quick-set emulsions.

Other Materials

It is almost always necessary to add a small amount of mineral filler such as hydrated lime, limestone dust, Portland cement, or fly ash to aid in stabilizing and setting the slurry. Water used in the slurry should be potable and compatible with the mix.

Mix Design

For slurry seal mix design, the following sources are recommended: ASTM D 3910 "Standard Practices for Design, Testing and Construction of Slurry Seal" or "Recommended Performance Guidelines for Emulsified Asphalt Slurry Seal Surfaces," available from the International Slurry Surfacing Association.

Construction Process

Slurry Seal Machine

The machine used for production of the slurry seal is a self-contained, continuous-flow mixing unit (see Figure 12.22). It is capable of accurately delivering predetermined amounts of aggregate, mineral filler (if required), water, and asphalt emulsion to the mixing chamber. It also discharges the thoroughly mixed materials onto the prepared surface. Certain basic features are common to all batch-type slurry machines. They are truck-mounted units with separate storage tanks, bins, and metering systems for emulsified asphalt, water, aggregate, and mineral filler. The slurry machine has a continuous-flow mixing unit, either a single or double pugmill, from which the slurry is discharged into a spreader box. The box is equipped with flexible squeegees and a device for adjustable placement width. Spreader boxes may be equipped with hydraulically powered augers to keep the slurry in motion and help keep the mixture uniformly spread across the spreader box width. These are helpful when quick-setting emulsion is used. One type of slurry mixer unit is depicted in the schematic drawing (Figure 12.23); continuously self-loading machines capable of placing many miles (kilometers) per day of coarse slurry are used in many regions of the United States.

The slurry is usually applied in a thickness of ⅛–¼ inch (3–6 millimeters). It comes directly from a traveling mixing plant into an attached spreader box that spreads the slurry by a squeegee-type action.

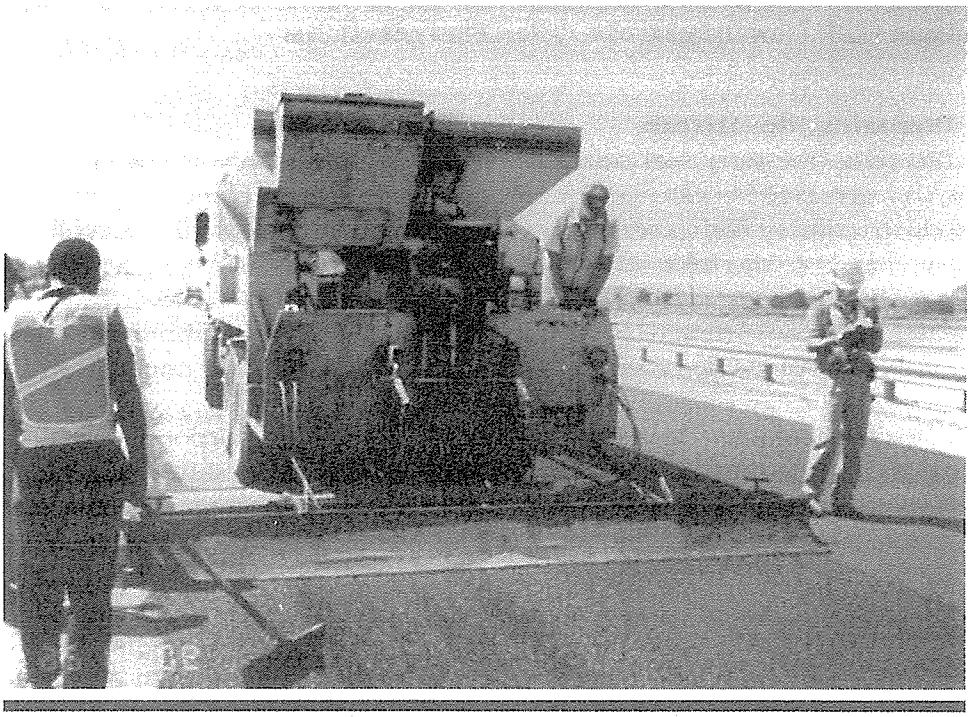


Figure 12.22 Slurry Seal Machine

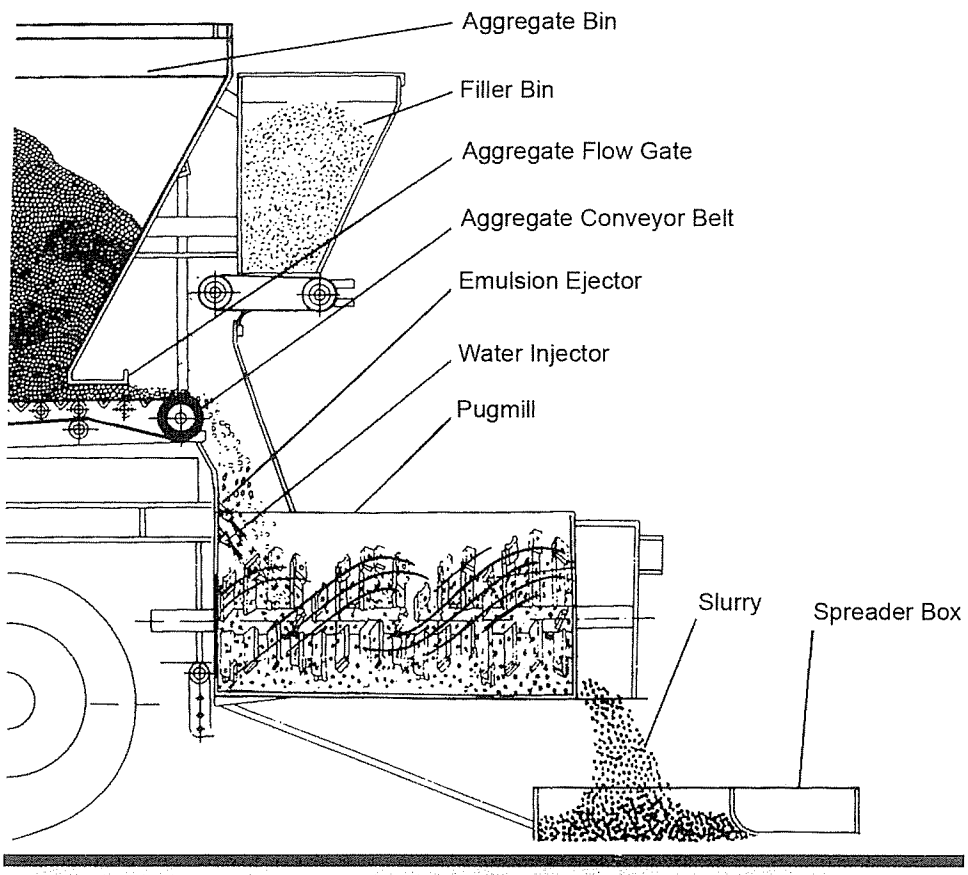


Figure 12.23 Flow Diagram of a Typical Slurry Seal Mixer

Preparing the Mixture

Blending the slurry seal materials in varying proportions in the laboratory is a great aid in selecting the proper mixture. Correct blending should produce a slurry with a creamy texture that will flow smoothly in a rolling wave ahead of the strike-off squeegee. This slurry should be a semifluid, homogeneous mass with no emulsion runoff.

After mix proportions have been determined in the laboratory, it is advisable to place one or more trial mixes. This should be done either at the job site or in a location where small spreads of the slurry seal would not be objectionable. The trial sections serve a two-fold purpose:

- To calibrate the feeding and metering devices on the slurry machine. Aggregate flow should be determined for different gate openings and the amount of emulsion pumped per revolution of the aggregate feed belt.
- To ensure that the slurry mix proportions are right. It is often necessary to make several trial runs to find the best blend of materials—even when starting with a laboratory-determined mixture. Always use exactly the same materials in the laboratory design work that are to be used for the field construction.

If lumping, balling, or unmixed aggregate is observed, the slurry should be removed from the pavement. It should also be removed if coarser aggregate particles settle to the bottom of the mix. Streaks, such as those caused by oversized aggregate, should be repaired at once with a hand squeegee.

Placing the Mixture

Just before applying the slurry, the pavement surface should be cleaned of all dirt, dust, mud spots, vegetation, and other foreign matter. A tack coat of diluted emulsified asphalt of the same type and grade specified for the slurry may be required directly ahead of the slurry application. With relatively new asphalt pavements, the tack coat may be omitted. The pavement surface should always be damp but with no free water in front of the slurry machine.

It is especially important to get a homogenous mix, one that will produce a slurry with a creamy texture that will flow smoothly in a rollaway wave inside the spreader box. A non-homogenous mixture will cause an asphalt-rich surface and many ensuing problems.

Special care must be taken with longitudinal and transverse joints to prevent excessive buildup of slurry (ridging) and streaking. It is best to make the joint after the first placed lane is either completely cured or is still in a semifluid condition. For good appearance and durability, a joint should not be made when the lane to be joined is only partially set because tearing and scarring might result.

Quite often a canvas or burlap drag is pulled behind the spreader box to improve the joint and overall surface appearance. Drags should be changed regularly. Hand squeegees and hand drags are used to improve joints and place the slurry in areas inaccessible to the machine.

On flat grades, slurry in the spreader box must be kept at a uniform thickness. On high-crowned pavements or superelevated curves, slurry should be diverted to the high side of the spreader box. Gravity will keep the low side filled. Spreading slurry in hilly areas is easier if the slurry machine travels in an uphill direction. If circumstances require placement in a downhill direction, the slurry must be thickened to prevent it from flowing ahead of the machine.

Compaction

Rolling slurry seal is only needed in those areas where pneumatic-tire rolling will improve durability. Such areas include taxiways, runways, truck terminal yards, and intersections of heavily traveled roads. All of these are subject to power-steering turns, braking, or acceleration forces. For rolling, a 5-ton (4500 kilograms) pneumatic roller with 50 pounds per square inch (345 kiloPascals) tire pressure will be most effective. Rolling can start as soon as clear water can be pressed out of the slurry mixture with a piece of paper without discoloring the paper. In most cases, traffic will iron out the slurry and close any hairline cracks of dehydration. Rolling usually is not needed unless the thickness is more than ¼ inch (6 millimeters) or if late season work is involved.

Curing/Traffic Control

Slurry should be placed only when the temperature is at least 50°F (10°C) and rising and when no rain is expected. Roads with newly placed slurry should not be opened to traffic until the slurry has completely cured. As with rolling, traffic generally can be allowed on the slurry as soon as clear water can be pressed out of the slurry mixture with a piece of paper without discoloring the paper. The traffic, of course, must be controlled somewhat because quick stops or accelerations and the turning of a parked vehicle's wheels will damage the slurry.

12.4.5 Microsurfacing

Microsurfacing is similar to slurry seal but with several exceptions. First, a polymer-modified binder is used. Second, because a polymer-modified binder makes a much stiffer mix, the mixture can be placed more than one aggregate in thickness; however, the thickness is limited to about ¾ inch. Again, because of the mix stiffness, a more substantial mixer is required, and twin-shafted paddles or spiral augers are required to provide a uniform flow of the mix in the spreader box.

Microsurfacing consists of a polymer-modified emulsion, a high-quality aggregate, mineral filler, additives, and water. It is generally required that the mixture include 82–90 percent aggregate and the following as a percentage of the dry aggregate:

- 1.5–3.0 percent Portland cement as a mineral filler
- 5.5–9.5 percent residual asphalt binder

Materials

The emulsified asphalt is a quick-setting, polymer-modified asphalt emulsion. The aggregates used for microsurfacing should meet the requirements of Type II and Type III aggregates as set forth in section 12.4.4. The mineral filler is a non-air entrained Portland cement or hydrated lime that is free from lumps. The type and amount of mineral filler is determined by a laboratory mix design. An increase or decrease of less than 1 percent may be required when the microsurfacing is being placed to provide better consistency or set times. Materials evaluation and mix design should be done in accordance with the International Slurry Surfacing Association's "Recommended Performance Guide for Micro-Surfacing, A143" (latest version).

Equipment

Mixing Equipment

The microsurfacing is mixed in a continuous-flow mixing unit (see Figure 12.24) that can accurately deliver and proportion the aggregate, emulsified asphalt, mineral filler, control-setting additive, and water to a revolving multi-blade, double-shafted mixer, and to discharge the mixed microsurfacing on a continuous-flow basis.

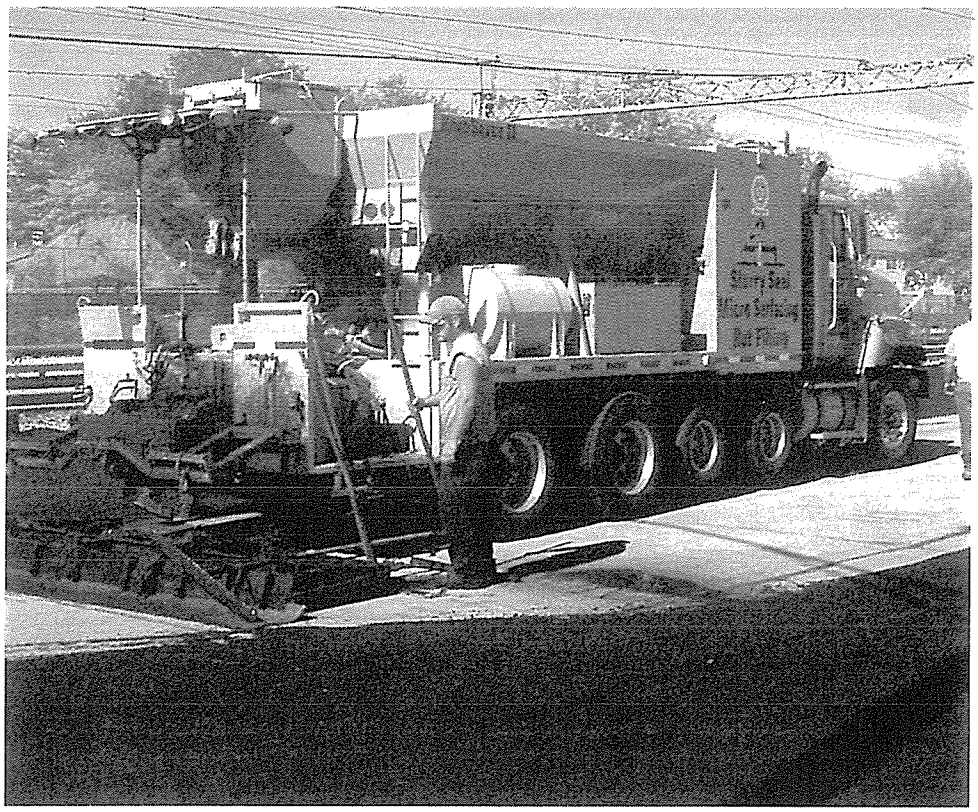


Figure 12.24 Microsurfacing Machine

Spreading Equipment

The mixture is spread uniformly in the surfacing box (see Figure 12.25) by means of twin-shafted paddles or spiral augers fixed in the spreader box. A front seal is provided to ensure no loss of the mixture at the road contact point. A rear seal acts as a final strike-off. The spreader box and rear strike-off are designed and operated to achieve a uniform consistency and to produce a free flow of material to the rear strike-off.

One feature of microsurfacing is that it can be used to fill ruts or channels in the traffic wheelpaths, provided the pavement is no longer in a plastic flow condition. It can be used to fill ruts, utility cuts, depressions in the existing surface, etc. Ruts of ½ inch (12.7 millimeters) or greater in depth can be filled independently with a rut-filling spreader box (see Figure 12.26) either 5 feet (1.5 meters) or 6 feet (1.8 meters) in width. For irregular or shallow rutting of less than ½ inch (12.5 millimeters) in depth, a full-width scratch-coat pass may be used. Ruts that exceed a depth of 1½ inches (37.5 millimeters) may require multiple placements with the rut-filling spreader box to restore the cross-section. All rut-filling material should cure under traffic for at least 24 hours before additional material is placed.



Figure 12.25 Microsurfacing Spreader Box

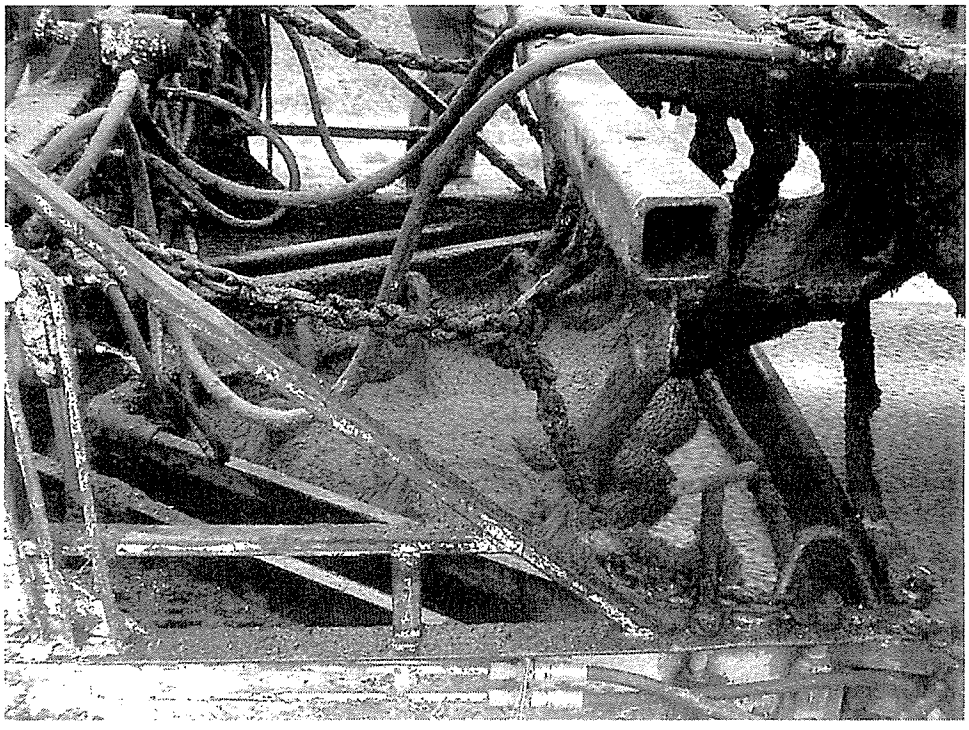


Figure 12.26 Rut-Filling Box

Weather Limitations

Microsurfacing should not be applied if either the pavement or air temperature is below 50°F (10°C) and falling, but may be applied when both pavement and air temperatures are above 45°F (7°C) and rising. No microsurfacing should be applied when the possibility exists that the finished product will freeze within 24 hours.

Surface Preparation

Before microsurfacing is placed, the surface of the existing pavement should be cleaned and prepared. Tight cracks should be cleaned and sealed, wide cracks should be filled, and the pavement should be thoroughly broomed and cleaned to remove any loose dirt and contaminants. Any service entrances or driveways should be covered.

A tack coat is not usually applied for HMA pavements but can be required in specific applications. An asphalt emulsion—usually a diluted CSS or SS emulsion—is applied at a rate of 0.05–0.10 gallons/square yard (0.19–0.38 liters/square meter)

Application

It is recommended that a test strip be placed in conditions similar to those expected during the project. Pre-wetting the surface by applying a light fog ahead of the spreader box is effective when the weather is hot or the pavement texture is rough. The water keeps the emulsion from prematurely breaking before the mix can be spread and finished, and aids in the penetration and sealing of small surface cracks by the emulsion.

Joints

No excess buildup, uncovered areas, or unsightly appearance should be permitted on longitudinal or transverse joints. The goal is to produce a minimum number of longitudinal joints throughout the project. When possible, longitudinal joints should be placed on lane lines. Half passes and odd-width passes should be used only in minimal amounts. If half passes are used, they should not be the last passes of any paved area. A maximum of 3 inches (75 millimeters) should be provided for the overlap of longitudinal lane line joints. Also, the joint should have no more than a ¼ inch (6 millimeters) difference in elevation when measured by placing a 10-foot (3 meters) straightedge over the joint and measuring the elevation drop-off.

12.4.6 Cape Seal

The term Cape seal is attributed to the Cape Provincial Administration of South Africa. A Cape seal can be defined as a single-layer surface treatment (chip seal) followed by an emulsion-mix slurry seal.

In order to have a successful Cape seal project, beyond following standard surface treatment and slurry seal specifications and methods, it is important that the surface treatment be placed as a single course only. The most critical element to avoid in a Cape seal is an excess of slurry, which can destroy the desired knobby surface texture (see Table 12.6).

Table 12.6
Quantities of Asphalt and Aggregate per Square Yard (Square Meter) for Cape Seal

½ in (12.5 mm) thick	gal/yd²-Asphalt (l/m²)	lb/yd² -Aggregate (kg/m²)	lb/yd² -Slurry Mixture (kg/m²)
Emulsion (RS-2, CRS-2)	0.30–0.45 (1.4–2.0)		
Cover aggregate			
ASTM or AASHTO			
Size No. 7			
½ in–No. 4		25–30	
(12.5–4.75 mm)		(14–16)	
Slurry seal			
Type I			6–10
			(3–5.5)

A cure time of four to ten days between placement of the surface treatment and subsequent slurry seal application should also be provided, during which time regular brooming should be done to remove loose cover material or other foreign material that would prevent the adherence of the slurry.

12.4.7 Sand Seal

A sand seal is an application of sprayed asphalt material covered with fine aggregate. It may be used to improve the skid resistance of slippery pavements or to seal against air and water infiltration. Construction procedures for sand seals are similar to those for single-surface treatments. Typically, asphalt is sprayed onto the surface to be treated, followed by an application of sand.

Usually, emulsion grades RS-1, CRS-1, MS-1, or HFMS-1 are used. They are applied at a rate of 0.15–0.20 gallons/square yard (0.70–0.90 liters/square meter). This is followed by 10–15 pounds/square yard (5.5–8 kilograms/square meter) of sand, or screenings cover.

12.5 Thin Overlays (Maintenance Blanket)

A thin asphalt surfacing—a thickness of 1–1½ inches (25–37.5 millimeters)—can be used to extend the life of an asphalt pavement. This use of a thin asphalt surfacing may also improve the ride quality and correct surface defects such as low skid resistance. Many times, a thin overlay is used in combination with milling operations (discussed in Chapter 13, "Rehabilitation Methods for Asphalt Pavements"). This type of surfacing is very applicable for use on residential streets. Milling may be required to maintain a minimum curb reveal.

A thin overlay should be used when preventive maintenance is the goal. Existing structural damage (fatigue cracking) should be repaired before the overlay, and the cause of any rutting (greater than ¼ inch) should be determined and corrected. Pavement layers that have rutting problems should be removed and replaced with materials that have sufficient stability to withstand the traffic loads.

The pavement materials and thickness should be chosen on the basis of the traffic and the structural needs of the pavement. The thickness should be a minimum of three times the nominal maximum aggregate size (as defined in Chapter 4, "Hot Mix Asphalt Mix Design"). Thus, for a pavement with a nominal maximum aggregate size of ½ inch (12.5 millimeters), the thickness should be at least 1½ inches (37.5 millimeters).

12.6 Guidance on Maintenance Procedures

The following is a summary of the appropriate maintenance procedures for various distress types. A detailed description of each distress type is provided in Chapter 11, "Pavement Management."

Distress Description		Repair Technique	Section Where Repair Technique Is Described
Alligator cracking		Full-depth patch	12.2.2
Bleeding or flushing asphalt		Cold mill and thin overlay	13.3
Corrugations (wash-boarding)		Deep or full-depth patch	12.2.2
Cracking		Crack seal or fill	12.3.3
Raveling	Light	Fog seal	12.4.2
	Medium	Fog seal, slurry seal, or microsurfacing	12.4.2, 12.4.7
	Severe	Slurry seal, microsurfacing, or thin overlay	12.4.2, 12.4.7, 12.5
Grade depressions		Cold mill and overlay	13.3
Longitudinal cracking		Crack seal or fill	12.3.3
Low skid resistance		Seal coat or slurry seal	12.4.2, 12.4.3
Polished aggregate		Seal coat or slurry seal	12.4.2, 12.4.3
Potholes		Injection or full-depth patch	12.2.4
Reflection cracking		Crack seal or fill	12.3.3
Rutting		Cold mill and overlay	13.3
Slippage cracking		Deep or full-depth patch	12.2.2
Transverse cracks		Crack seal or fill	12.3.3
Upheaval		Full-depth patch	12.2.2

Resources

1. MS-16, *Asphalt in Pavement Maintenance.* The Asphalt Institute, Lexington, KY.
2. MS-19, *Basic Asphalt Emulsion Manual.* The Asphalt Institute, Lexington, KY.
3. Gransberg, Douglas and James, David. *Chip Seal Best Practices,* NCHRP Synthesis 342. Transportation Research Board, 2005.
4. *FHWA-IF-00–027, Selecting a Pavement Maintenance Treatment for Flexible Pavements.* August 2000.
5. *FHWA NHI Course No. 13154, Pavement Preservation, The Preventive Maintenance Concept.* September 1999.
6. MS-22, *HMA Construction.* The Asphalt Institute, Lexington, KY.